

The empirical recurrence for OEIS A236898: recovering a conjecture that is not written down

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Abstract

OEIS A236898 records “Empirical recurrence of order 86”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 465$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 86, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 86 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236898 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with the maximum plus the upper median of every 2×2 subblock equal.”. It has offset 1 and begins

6545, 337861, 13845208, 715994489, 35006442815, 1767780514348, 88420063405403, 4442899040820873, ...

The entry states:

Empirical recurrence of order 86 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:39:35, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 465$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 465$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 930$ exact terms of the model returns order 86 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{86} c_i a(n-i),$$

c_1	73	c_{23}	7861764114554653299710997	c_{45}	−608640657586578401
c_2	−77	c_{24}	26460816568911828397260101	c_{46}	4726159843962887073
c_3	−81775	c_{25}	−200100866808425288438264648	c_{47}	2100289830971248852
c_4	985926	c_{26}	−469527542300916294588581489	c_{48}	−1852654598959234117
c_5	35901474	c_{27}	4093772117112906693549269047	c_{49}	−6131136060511647790
c_6	−645839894	c_{28}	6598237900786439061054302612	c_{50}	60003950193394560310
c_7	−7863156887	c_{29}	−68238598350203806326364989641	c_{51}	15112414066539731742
c_8	200464514037	c_{30}	−72487863772881012204267724568	c_{52}	−1616042119288908042
c_9	848648780450	c_{31}	936478081940283534395430235642	c_{53}	−3136396434045602693
c_{10}	−37077510976621	c_{32}	598495476724848691492767165593	c_{54}	36291890036749582152
c_{11}	−16627417621963	c_{33}	−10667578290493323584992460509426	c_{55}	54593954263455580378
c_{12}	4528352453656820	c_{34}	−3256417720260063955849614313003	c_{56}	−6798032449578843065
c_{13}	−7825975928710845	c_{35}	101509125205262344943858379568414	c_{57}	−7929675165426965534
c_{14}	−388239608280621424	c_{36}	3698455503963602840398741859872	c_{58}	10604572235535551058
c_{15}	1225032854300446672	c_{37}	−810882554070627982288747659946188	c_{59}	95478904841735160173
c_{16}	24342818427261924926	c_{38}	149249356500952684548651893947294	c_{60}	−13732867585293486829
c_{17}	−102296329813947324093	c_{39}	5458013554117973190013766805417016	c_{61}	−9450398024213578452
c_{18}	−1149159411990632719418	c_{40}	−2008565635085255608336729785621968	c_{62}	14694391410469723104
c_{19}	5813937557680945027573	c_{41}	−31036906827801000494089242844886468	c_{63}	76066591101071137445
c_{20}	41719461851781929593258	c_{42}	16263201986608714725677474146542800	c_{64}	−12911062685462144754
c_{21}	−243796164618304604118856	c_{43}	149351983303895132990484958167968096	c_{65}	−4909165330620297092
c_{22}	−1182809358012050815910448	c_{44}	−98347302497683369700007946085044248	c_{66}	92422540615877849562

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 86, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $86 + 4$ terms removed returns order 86 as well, so $\deg q_{\min} = 86 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $86 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 86 that this sequence satisfies — and that family turns out to have exactly one member.

References

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